

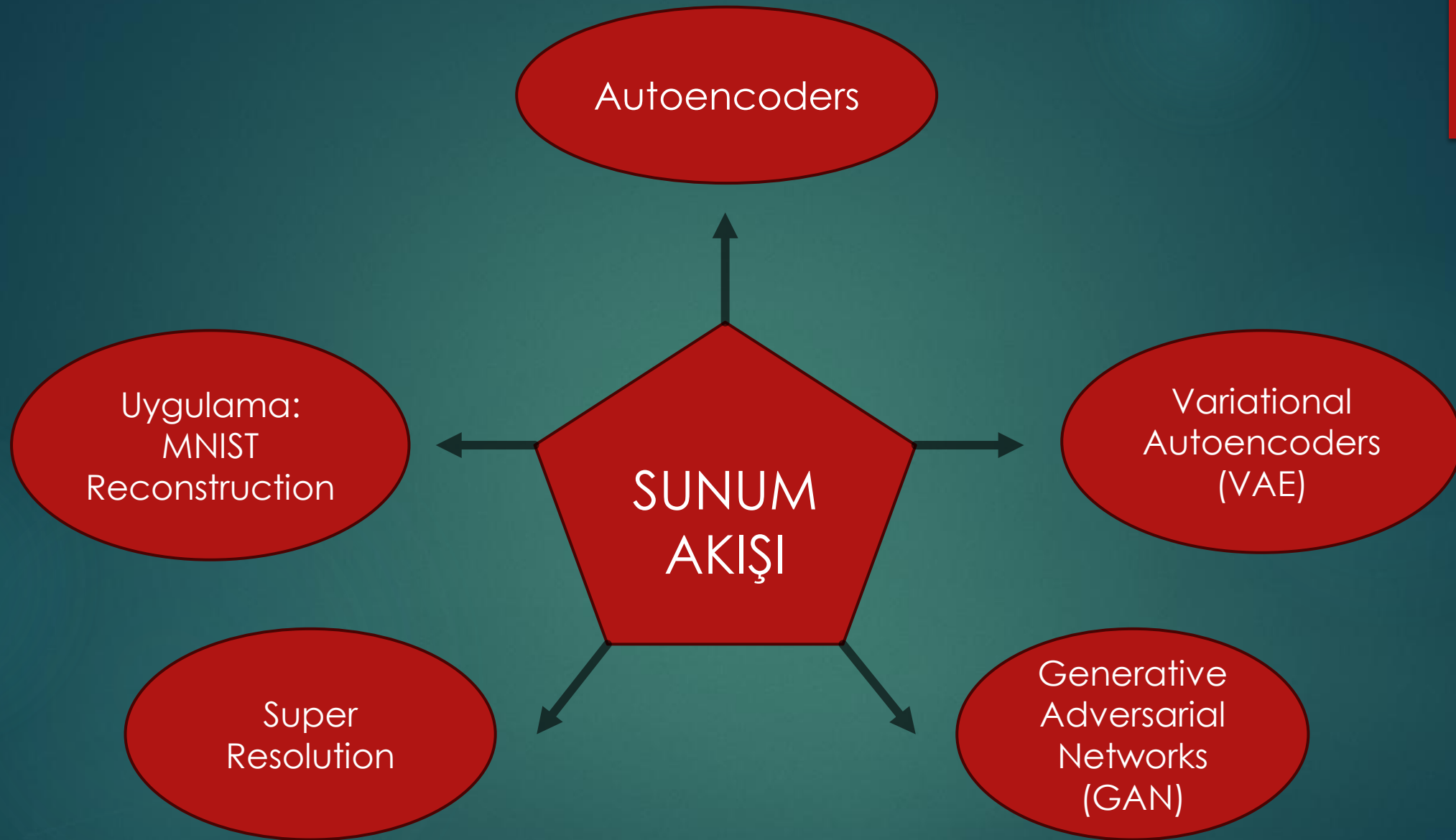
SUNUM 11

Üretken Modeller ve Autoencoders

AD: MEHMET EMRE ŞENGÖZ

NO:2322705065





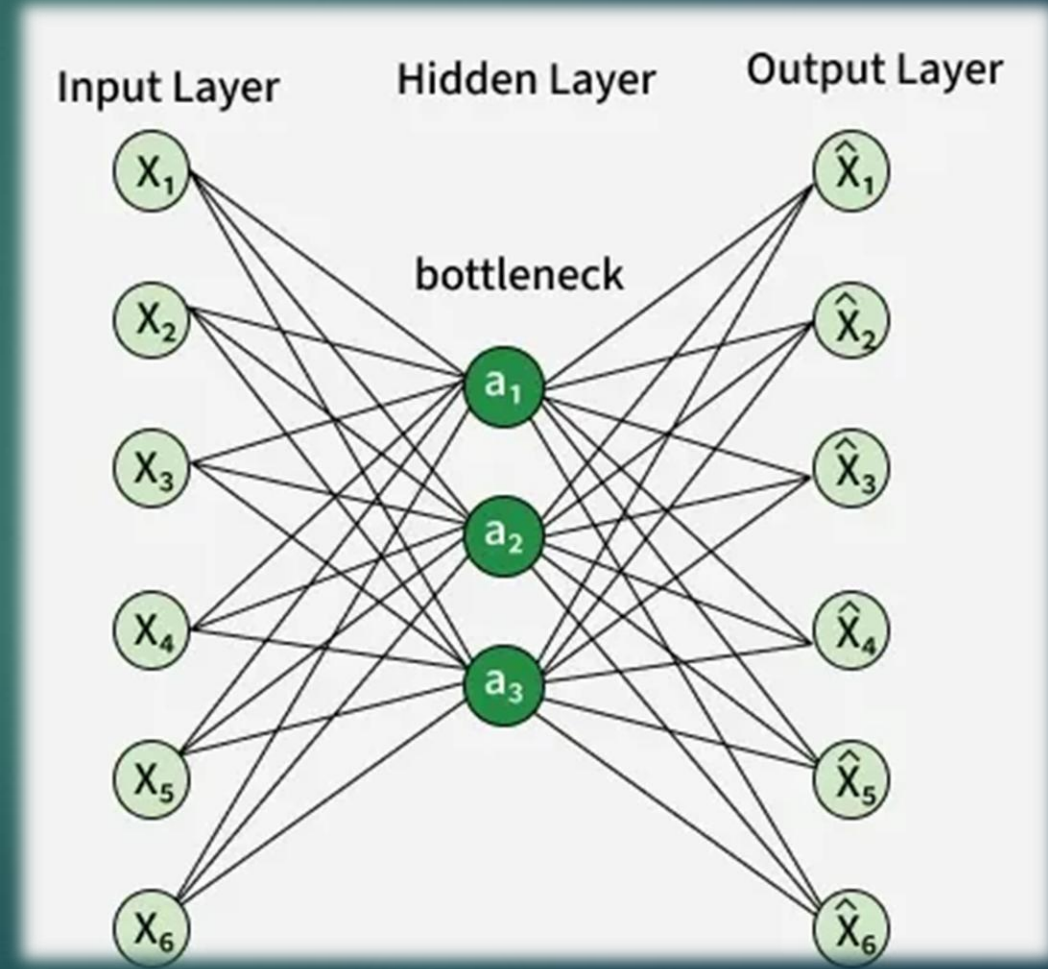
Autoencoders Nedir?

1) ENCODER

2) BOTTLENECK

3) DECODER

NEDEN İHTİYAÇ ?



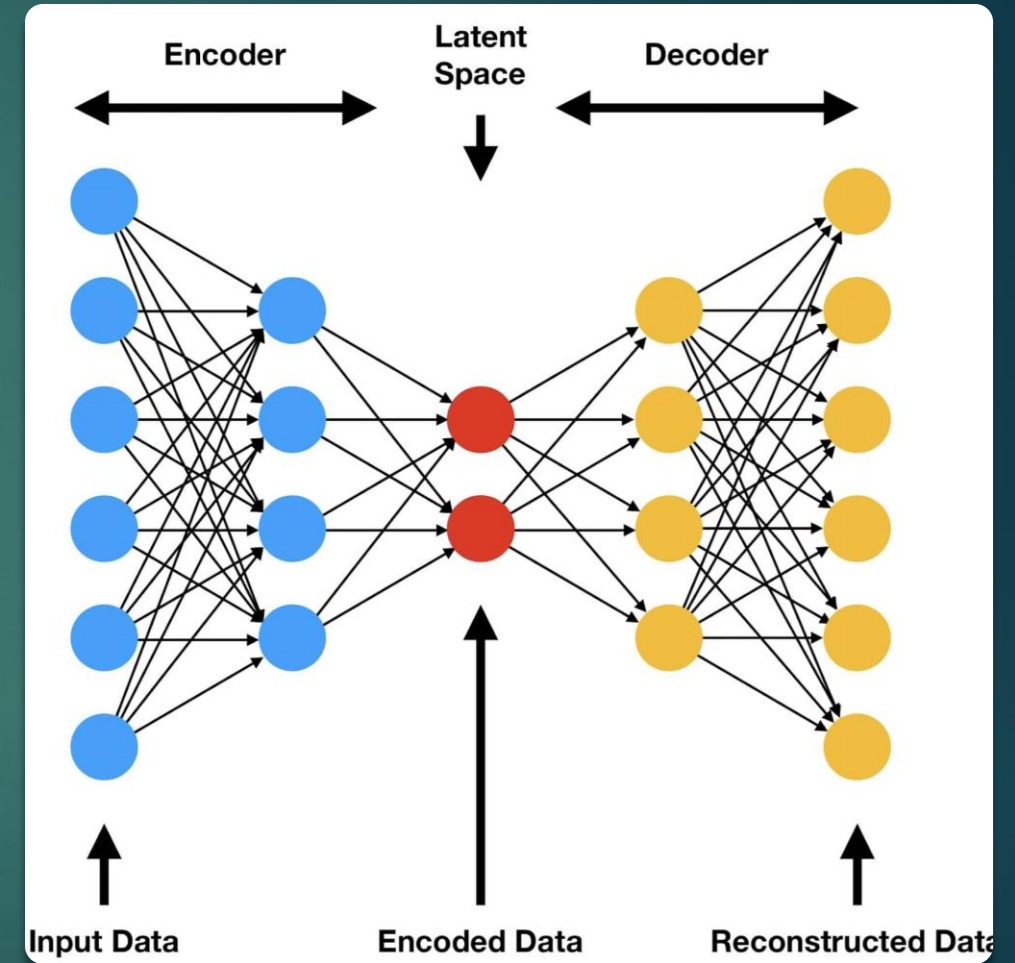
Latent Space

→ Nedir ?

→ 28x28 (MNIST)

→ 784 → 32

→ Bottle Neck



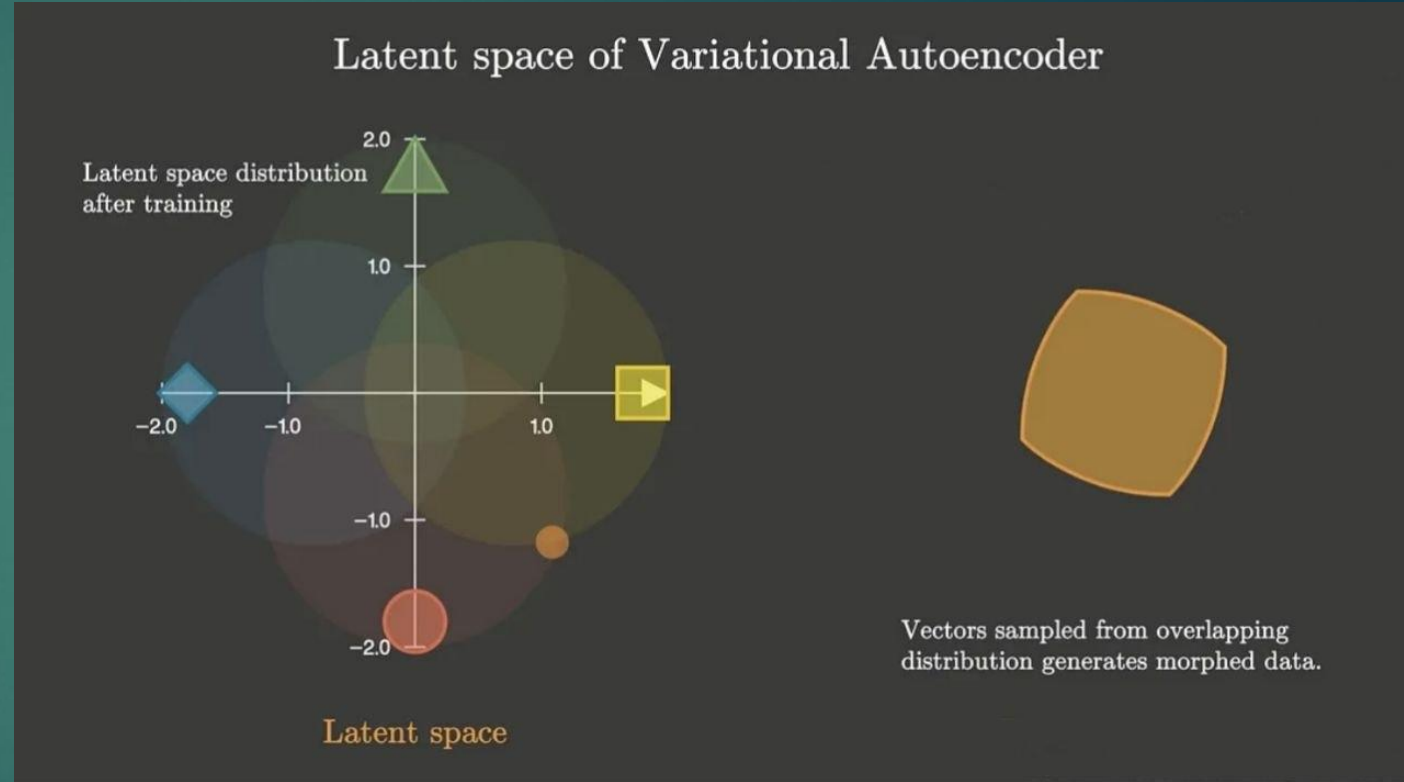
Variational Autoencoders (VAE)

→ Nedir ?

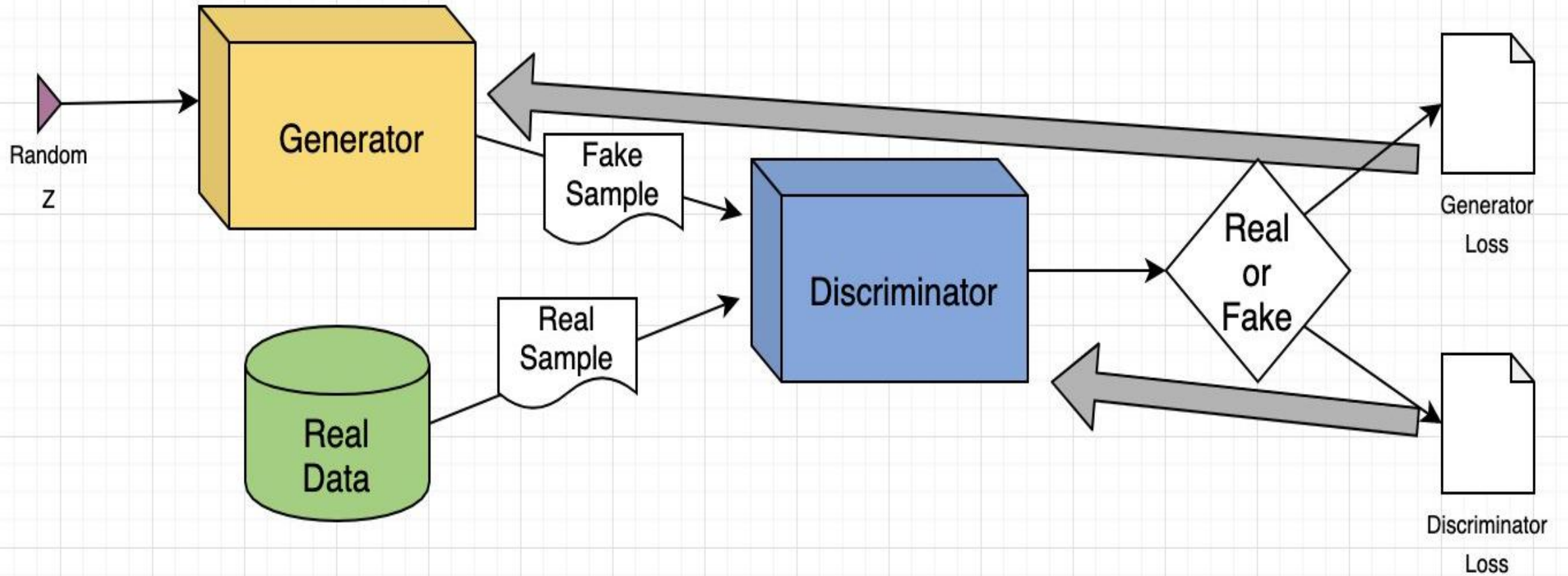
→ Olasılıksal Yaklaşım

→ Sürekli Gizli Uzay

→ Üretken Yetenek (Generative)



Generative Adversarial Networks (GAN)



GAN Uygulama Örnekleri



Super Resolution

- ➔ Nedir ?
- ➔ Piksel Tahmini
- ➔ Kullanım Alanları



Uygulama - MNIST Autoencoder

Model: "functional"

Layer (type)	Output Shape	Param #
input_layer (InputLayer)	(None, 784)	0
dense (Dense)	(None, 128)	100,480
dense_1 (Dense)	(None, 32)	4,128
dense_2 (Dense)	(None, 128)	4,224
dense_3 (Dense)	(None, 784)	101,136

Total params: 209,968 (820.19 KB)

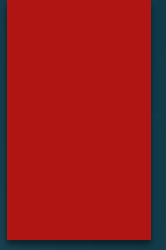
Trainable params: 209,968 (820.19 KB)

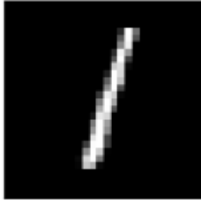
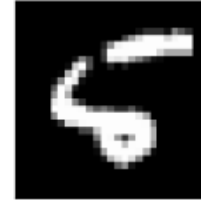
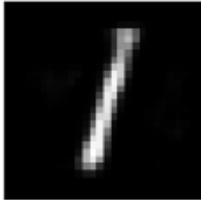
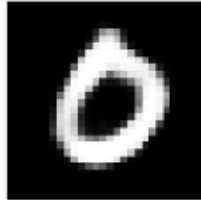
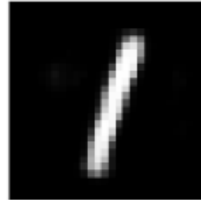
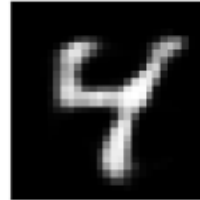
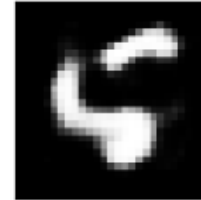
Non-trainable params: 0 (0.00 B)

Eğitim Süreci ve Parametreler

```
Epoch 1/10
235/235 ————— 2s 5ms/step - loss: 0.0606 - val_loss: 0.0327
Epoch 2/10
235/235 ————— 1s 5ms/step - loss: 0.0259 - val_loss: 0.0208
Epoch 3/10
235/235 ————— 1s 5ms/step - loss: 0.0192 - val_loss: 0.0170
Epoch 4/10
235/235 ————— 1s 6ms/step - loss: 0.0164 - val_loss: 0.0149
Epoch 5/10
235/235 ————— 1s 5ms/step - loss: 0.0147 - val_loss: 0.0137
Epoch 6/10
235/235 ————— 1s 5ms/step - loss: 0.0136 - val_loss: 0.0128
Epoch 7/10
235/235 ————— 1s 5ms/step - loss: 0.0129 - val_loss: 0.0123
Epoch 8/10
235/235 ————— 1s 4ms/step - loss: 0.0124 - val_loss: 0.0119
Epoch 9/10
235/235 ————— 1s 5ms/step - loss: 0.0120 - val_loss: 0.0115
Epoch 10/10
235/235 ————— 1s 5ms/step - loss: 0.0117 - val_loss: 0.0112
Eğitim tamamlandı!
```

Final Sonuçları



Orijinal	Orijinal	Orijinal	Orijinal	Orijinal	Orijinal	Orijinal	Orijinal	Orijinal	Orijinal
									
Sonuç	Sonuç	Sonuç	Sonuç	Sonuç	Sonuç	Sonuç	Sonuç	Sonuç	Sonuç
									



TEŞEKKÜRLER